Theory and Examples

For Exercises 55 and 56 evaluate each limit by first converting each to a derivative at a particular x-value.

55. $\lim_{x \to 1} \dfrac{x^{50} - 1}{x - 1}$

56. $\lim_{x \to -1} \dfrac{x^{2/9} - 1}{x + 1}$

57. Find the value of a that makes the following function differentiable for all x-values.

$$g(x) = \begin{cases} ax, & \text{if } x < 0 \\ x^2 - 3x, & \text{if } x \geq 0 \end{cases}$$

58. Find the values of a and b that make the following function differentiable for all x-values.

$$f(x) = \begin{cases} ax + b, & x > -1 \\ bx^2 - 3, & x \leq -1 \end{cases}$$

59. The general polynomial of degree n has the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where $a_n \neq 0$. Find $P'(x)$.

3.5 Derivatives of Trigonometric Functions

Derivative of the Sine Function

To calculate the derivative of $f(x) = \sin x$, for x measured in radians, we combine the limits in Example 5a and Theorem 7 in Section 2.4 with the angle sum identity for the sine function:

$$\sin(x + h) = \sin x \cos h + \cos x \sin h.$$

If $f(x) = \sin x$, then

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \to 0} \frac{\sin(x + h) - \sin x}{h} \qquad \text{Derivative definition}$$

$$= \lim_{h \to 0} \frac{(\sin x \cos h + \cos x \sin h) - \sin x}{h} \qquad \text{Identity for } \sin(x + h)$$

$$= \lim_{h \to 0} \frac{\sin x (\cos h - 1) + \cos x \sin h}{h}$$

$$= \lim_{h \to 0} \left(\sin x \cdot \frac{\cos h - 1}{h} \right) + \lim_{h \to 0} \left(\cos x \cdot \frac{\sin h}{h} \right)$$

$$= \sin x \cdot \underbrace{\lim_{h \to 0} \frac{\cos h - 1}{h}}_{\text{limit } 0} + \cos x \cdot \underbrace{\lim_{h \to 0} \frac{\sin h}{h}}_{\text{limit } 1}$$

$$= \sin x \cdot 0 + \cos x \cdot 1 = \cos x. \qquad \text{Example 5a and Theorem 7, Section 2.4}$$

The derivative of the sine function is the cosine function:

$$\frac{d}{dx}(\sin x) = \cos x.$$

Derivative of the Cosine Function

With the help of the angle sum formula for the cosine function,

$$\cos(x + h) = \cos x \cos h - \sin x \sin h,$$

we can compute the limit of the difference quotient:

$$\begin{aligned}
\frac{d}{dx}(\cos x) &= \lim_{h\to 0}\frac{\cos(x+h)-\cos x}{h} && \text{Derivative definition}\\
&= \lim_{h\to 0}\frac{(\cos x\cos h-\sin x\sin h)-\cos x}{h} && \text{Cosine angle sum identity}\\
&= \lim_{h\to 0}\frac{\cos x(\cos h-1)-\sin x\sin h}{h}\\
&= \lim_{h\to 0}\left(\cos x\cdot\frac{\cos h-1}{h}\right)-\lim_{h\to 0}\left(\sin x\cdot\frac{\sin h}{h}\right)\\
&= \cos x\cdot\lim_{h\to 0}\frac{\cos h-1}{h}-\sin x\cdot\lim_{h\to 0}\frac{\sin h}{h} && \text{Example 5a and Theorem 7, Section 2.4}\\
&= \cos x\cdot 0-\sin x\cdot 1\\
&= -\sin x.
\end{aligned}$$

The derivative of the cosine function is the negative of the sine function:

$$\frac{d}{dx}(\cos x) = -\sin x.$$

Derivatives of the Other Basic Trigonometric Functions

Because $\sin x$ and $\cos x$ are differentiable functions of x, the related functions

$$\tan x = \frac{\sin x}{\cos x}, \qquad \cot x = \frac{\cos x}{\sin x}, \qquad \sec x = \frac{1}{\cos x}, \qquad \text{and} \qquad \csc x = \frac{1}{\sin x}$$

are differentiable at every value of x at which they are defined. Their derivatives, calculated from the Quotient Rule, are given by the following formulas. Notice the negative signs in the derivative formulas for the cofunctions.

The derivatives of the other trigonometric functions:

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

EXAMPLE 5 Find $d(\tan x)/dx$.

EXAMPLE 6 Find y'' if $y = \sec x$.

DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS

Recall the definition of the arcsine function:

$$y = \sin^{-1}x \qquad \text{means} \qquad \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leqslant y \leqslant \frac{\pi}{2}$$

Differentiating $\sin y = x$ implicitly with respect to x, we obtain

$$\cos y \frac{dy}{dx} = 1 \qquad \text{or} \qquad \frac{dy}{dx} = \frac{1}{\cos y}$$

Now $\cos y \geqslant 0$, since $-\pi/2 \leqslant y \leqslant \pi/2$, so

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

Therefore

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1 - x^2}}$$

$$\sec^2 y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}$$

EXAMPLE 5 Differentiate (a) $y = \dfrac{1}{\sin^{-1} x}$ and (b) $f(x) = x \arctan\sqrt{x}$.

DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS

$$\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}(\csc^{-1}x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2} \qquad \frac{d}{dx}(\cot^{-1}x) = -\frac{1}{1+x^2}$$

45–54 Find the derivative of the function. Simplify where possible.

45. $y = \tan^{-1}\sqrt{x}$

46. $y = \sqrt{\tan^{-1}x}$

47. $y = \sin^{-1}(2x + 1)$

48. $g(x) = \sqrt{x^2 - 1}\,\sec^{-1}x$

49. $G(x) = \sqrt{1 - x^2}\,\arccos x$

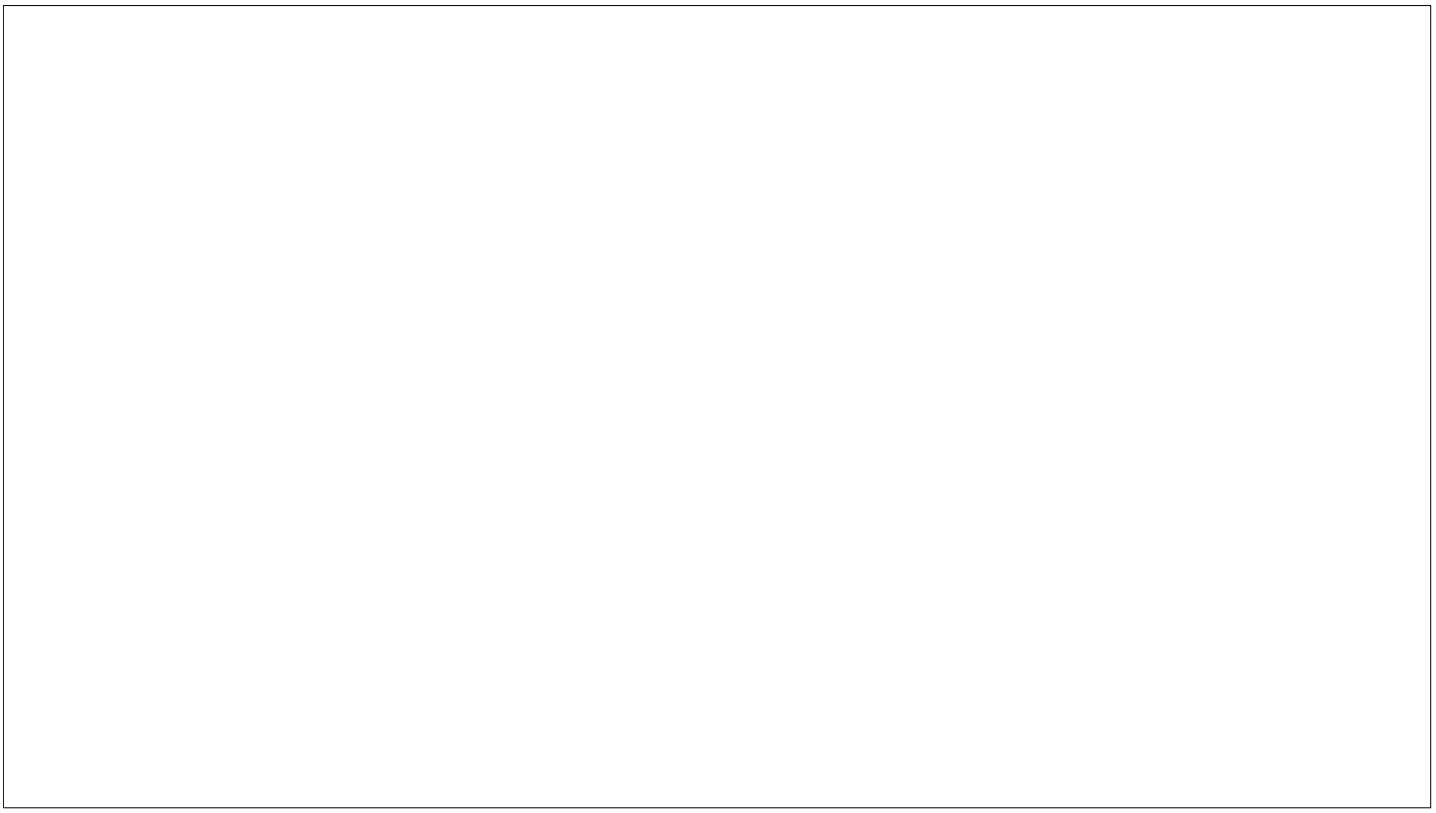
50. $y = \tan^{-1}\left(x - \sqrt{1 + x^2}\right)$

51. $h(t) = \cot^{-1}(t) + \cot^{-1}(1/t)$

52. $F(\theta) = \arcsin\sqrt{\sin\theta}$

53. $y = \cos^{-1}(e^{2x})$

54. $y = \arctan\sqrt{\dfrac{1 - x}{1 + x}}$



3.6 The Chain Rule

THEOREM 2—The Chain Rule If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x, then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x, and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

where dy/du is evaluated at $u = g(x)$.

EXAMPLE 2 An object moves along the x-axis so that its position at any time $t \geq 0$ is given by $x(t) = \cos(t^2 + 1)$. Find the velocity of the object as a function of t.

EXAMPLE 1 Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$.

EXAMPLE 2 Differentiate (a) $y = \sin(x^2)$ and (b) $y = \sin^2 x$.

4 THE POWER RULE COMBINED WITH THE CHAIN RULE If n is any real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$$

Alternatively,

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$$

EXAMPLE 3 Differentiate $y = (x^3 - 1)^{100}$.

EXAMPLE 4 Find $f'(x)$ if $f(x) = \dfrac{1}{\sqrt[3]{x^2 + x + 1}}$.

EXAMPLE 5 Find the derivative of the function

$$g(t) = \left(\frac{t - 2}{2t + 1} \right)^9$$

Repeated Use of the Chain Rule

We sometimes have to use the Chain Rule two or more times to find a derivative.

EXAMPLE 4 Find the derivative of $g(t) = \tan(5 - \sin 2t)$.

EXAMPLE 6 Differentiate $y = (2x + 1)^5(x^3 - x + 1)^4$.

EXAMPLE 7 Differentiate $y = e^{\sin x}$.

3.4 EXERCISES

1–6 Write the composite function in the form $f(g(x))$. [Identify the inner function $u = g(x)$ and the outer function $y = f(u)$.] Then find the derivative dy/dx.

1. $y = \sin 4x$
2. $y = \sqrt{4 + 3x}$
3. $y = (1 - x^2)^{10}$
4. $y = \tan(\sin x)$
5. $y = e^{\sqrt{x}}$
6. $y = \sin(e^x)$

7–46 Find the derivative of the function.

7. $F(x) = (x^4 + 3x^2 - 2)^5$
8. $F(x) = (4x - x^2)^{100}$
9. $F(x) = \sqrt[4]{1 + 2x + x^3}$
10. $f(x) = (1 + x^4)^{2/3}$
11. $g(t) = \dfrac{1}{(t^4 + 1)^3}$
12. $f(t) = \sqrt[3]{1 + \tan t}$
13. $y = \cos(a^3 + x^3)$
14. $y = a^3 + \cos^3 x$
15. $y = xe^{-kx}$
16. $y = 3\cot(n\theta)$
17. $g(x) = (1 + 4x)^5(3 + x - x^2)^8$
18. $h(t) = (t^4 - 1)^3(t^3 + 1)^4$
19. $y = (2x - 5)^4(8x^2 - 5)^{-3}$
20. $y = (x^2 + 1)\sqrt[3]{x^2 + 2}$

EXERCISES 3.6

Derivative Calculations

In Exercises 1–8, given $y = f(u)$ and $u = g(x)$, find $dy/dx = dy/dx = f'(g(x))g'(x)$.

1. $y = 6u - 9, \quad u = (1/2)x^4$

2. $y = 2u^3, \quad u = 8x - 1$

3. $y = \sin u, \quad u = 3x + 1$

4. $y = \cos u, \quad u = -\frac{x}{3}$

5. $y = \sqrt{u}, \quad u = \sin x$

6. $y = \sin u, \quad u = x - \cos x$

7. $y = \tan u, \quad u = \pi x^2$

8. $y = -\sec u, \quad u = \frac{1}{x} + 7x$

In Exercises 9–18, write the function in the form $y = f(u)$ and $u = g(x)$. Then find dy/dx as a function of x.

9. $y = (2x + 1)^5$

10. $y = (4 - 3x)^9$

11. $y = \left(1 - \frac{x}{7}\right)^{-7}$

12. $y = \left(\frac{\sqrt{x}}{2} - 1\right)^{-10}$

13. $y = \left(\frac{x^2}{8} + x - \frac{1}{x}\right)^4$

14. $y = \sqrt{3x^2 - 4x + 6}$

15. $y = \sec(\tan x)$

16. $y = \cot\left(\pi - \frac{1}{x}\right)$

17. $y = \tan^3 x$

18. $y = 5\cos^{-4} x$

25. $y = x^2 \sin^4 x + x \cos^{-2} x$ **26.** $y = \frac{1}{x}\sin^{-5} x - \frac{x}{3}\cos^3 x$

27. $y = \frac{1}{18}(3x - 2)^6 + \left(4 - \frac{1}{2x^2}\right)^{-1}$

28. $y = (5 - 2x)^{-3} + \frac{1}{8}\left(\frac{2}{x} + 1\right)^4$

29. $y = (4x + 3)^4(x + 1)^{-3}$ **30.** $y = (2x - 5)^{-1}(x^2 - 5x)^6$

31. $h(x) = x \tan\left(2\sqrt{x}\right) + 7$ **32.** $k(x) = x^2 \sec\left(\frac{1}{x}\right)$

33. $f(x) = \sqrt{7 + x \sec x}$ **34.** $g(x) = \frac{\tan 3x}{(x + 7)^4}$

35. $f(\theta) = \left(\frac{\sin\theta}{1 + \cos\theta}\right)^2$ **36.** $g(t) = \left(\frac{1 + \sin 3t}{3 - 2t}\right)^{-1}$

37. $r = \sin(\theta^2)\cos(2\theta)$ **38.** $r = \sec\sqrt{\theta}\tan\left(\frac{1}{\theta}\right)$

39. $q = \sin\left(\frac{t}{\sqrt{t + 1}}\right)$ **40.** $q = \cot\left(\frac{\sin t}{t}\right)$

In Exercises 41–58, find dy/dt.

41. $y = \sin^2(\pi t - 2)$ **42.** $y = \sec^2 \pi t$

43. $y = (1 + \cos 2t)^{-4}$ **44.** $y = (1 + \cot(t/2))^{-2}$

45. $y = (t \tan t)^{10}$ **46.** $y = (t^{-3/4} \sin t)^{4/3}$

47. $y = \left(\frac{t^2}{t^3 - 4t}\right)^3$ **48.** $y = \left(\frac{3t - 4}{5t + 2}\right)^{-5}$

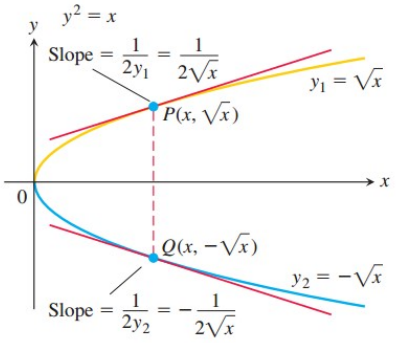
3.7 Implicit Differentiation

EXAMPLE 1

(a) If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

EXAMPLE 1 Find dy/dx if $y^2 = x$.



Implicit Differentiation

1. Differentiate both sides of the equation with respect to x, treating y as a differentiable function of x.
2. Collect the terms with dy/dx on one side of the equation and solve for dy/dx.

EXAMPLE 3 Find dy/dx if $y^2 = x^2 + \sin xy$ (Figure 3.30).

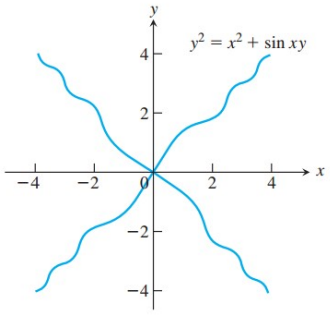


FIGURE 3.30 The graph of the equation in Example 3.

EXAMPLE 2

(a) Find y' if $x^3 + y^3 = 6xy$.

(b) Find the tangent to the folium of Descartes $x^3 + y^3 = 6xy$ at the point $(3, 3)$.

(c) At what points in the first quadrant is the tangent line horizontal?

EXAMPLE 3 Find y' if $\sin(x + y) = y^2 \cos x$.

Slopes, Tangents, and Normals

In Exercises 31–40, verify that the given point is on the curve and find the lines that are **(a)** tangent and **(b)** normal to the curve at the given point.

31. $x^2 + xy - y^2 = 1, \quad (2, 3)$

32. $x^2 + y^2 = 25, \quad (3, -4)$

33. $x^2y^2 = 9, \quad (-1, 3)$

34. $y^2 - 2x - 4y - 1 = 0, \quad (-2, 1)$

35. $6x^2 + 3xy + 2y^2 + 17y - 6 = 0, \quad (-1, 0)$

36. $x^2 - \sqrt{3}xy + 2y^2 = 5, \quad \left(\sqrt{3}, 2\right)$

37. $2xy + \pi \sin y = 2\pi, \quad (1, \pi/2)$

38. $x \sin 2y = y \cos 2x, \quad (\pi/4, \pi/2)$

39. $y = 2\sin(\pi x - y), \quad (1, 0)$

40. $x^2 \cos^2 y - \sin y = 0, \quad (0, \pi)$

3.8 Related Rates

Related Rates Equations

Suppose we are pumping air into a spherical balloon. Both the volume and radius of the balloon are increasing over time. If V is the volume and r is the radius of the balloon at an instant of time, then

$$V = \frac{4}{3}\pi r^3.$$

Using the Chain Rule, we differentiate both sides with respect to t to find an equation relating the rates of change of V and r,

$$\frac{dV}{dt} = \frac{dV}{dr}\frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

EXAMPLE 1 Water runs into a conical tank at the rate of 9 ft^3/min. The tank stands point down and has a height of 10 ft and a base radius of 5 ft. How fast is the water level rising when the water is 6 ft deep?

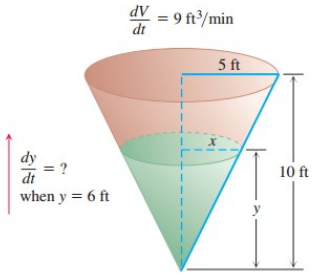


FIGURE 3.33 The geometry of the conical tank and the rate at which water fills the tank determine how fast the water level rises (Example 1).

Related Rates Problem Strategy

1. *Draw a picture and name the variables and constants*. Use t for time. Assume that all variables are differentiable functions of t.
2. *Write down the numerical information* (in terms of the symbols you have chosen).
3. *Write down what you are asked to find* (usually a rate, expressed as a derivative).
4. *Write an equation that relates the variables*. You may have to combine two or more equations to get a single equation that relates the variable whose rate you want to the variables whose rates you know.
5. *Differentiate with respect to t*. Then express the rate you want in terms of the rates and variables whose values you know.
6. *Evaluate*. Use known values to find the unknown rate.

EXAMPLE 2 A hot air balloon rising straight up from a level field is tracked by a range finder 150 m from the liftoff point. At the moment the range finder's elevation angle is $\pi/4$, the angle is increasing at the rate of 0.14 rad/min. How fast is the balloon rising at that moment?

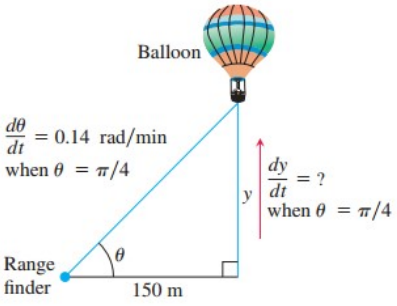


FIGURE 3.34 The rate of change of the balloon's height is related to the rate of change of the angle the range finder makes with the ground (Example 2).

EXAMPLE 3 A police cruiser, approaching a right-angled intersection from the north, is chasing a speeding car that has turned the corner and is now moving straight east. When the cruiser is 0.6 mi north of the intersection and the car is 0.8 mi to the east, the police determine with radar that the distance between them and the car is increasing at 20 mph. If the cruiser is moving at 60 mph at the instant of measurement, what is the speed of the car?

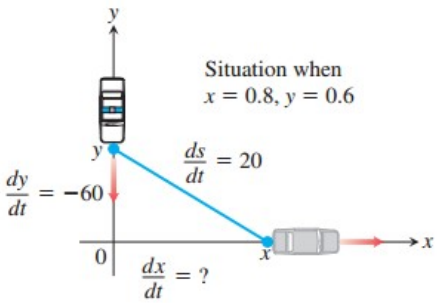


FIGURE 3.35 The speed of the car is related to the speed of the police cruiser and the rate of change of the distance s between them (Example 3).

EXERCISES 3.8

1. **Area** Suppose that the radius r and area $A = \pi r^2$ of a circle are differentiable functions of t. Write an equation that relates dA/dt to dr/dt.
2. **Surface area** Suppose that the radius r and surface area $S = 4\pi r^2$ of a sphere are differentiable functions of t. Write an equation that relates dS/dt to dr/dt.
3. Assume that $y = 5x$ and $dx/dt = 2$. Find dy/dt.
4. Assume that $2x + 3y = 12$ and $dy/dt = -2$. Find dx/dt.
5. If $y = x^2$ and $dx/dt = 3$, then what is dy/dt when $x = -1$?
6. If $x = y^3 - y$ and $dy/dt = 5$, then what is dx/dt when $y = 2$?
7. If $x^2 + y^2 = 25$ and $dx/dt = -2$, then what is dy/dt when $x = 3$ and $y = -4$?
8. If $x^2y^3 = 4/27$ and $dy/dt = 1/2$, then what is dx/dt when $x = 2$?
9. If $L = \sqrt{x^2 + y^2}$, $dx/dt = -1$, and $dy/dt = 3$, find dL/dt when $x = 5$ and $y = 12$.
10. If $r + s^2 + v^3 = 12$, $dr/dt = 4$, and $ds/dt = -3$, find dv/dt when $r = 3$ and $s = 1$.

11. If the original 24 m edge length x of a cube decreases at the rate of 5 m/min, when $x = 3$ m at what rate does the cube's

a. surface area change?

b. volume change?

12. A cube's surface area increases at the rate of 72 in^2/sec. At what rate is the cube's volume changing when the edge length is $x = 3$ in?

13. Volume The radius r and height h of a right circular cylinder are related to the cylinder's volume V by the formula $V = \pi r^2 h$.

a. How is dV/dt related to dh/dt if r is constant?

b. How is dV/dt related to dr/dt if h is constant?

c. How is dV/dt related to dr/dt and dh/dt if neither r nor h is constant?

14. Volume The radius r and height h of a right circular cone are related to the cone's volume V by the equation $V = (1/3)\pi r^2 h$.

a. How is dV/dt related to dh/dt if r is constant?

b. How is dV/dt related to dr/dt if h is constant?

c. How is dV/dt related to dr/dt and dh/dt if neither r nor h is constant?

b. How is ds/dt related to dy/dt and dz/dt if x is constant?

c. How are dx/dt, dy/dt, and dz/dt related if s is constant?

19. Area The area A of a triangle with sides of lengths a and b enclosing an angle of measure θ is

$$A = \frac{1}{2}ab\sin\theta.$$

a. How is dA/dt related to $d\theta/dt$ if a and b are constant?

b. How is dA/dt related to $d\theta/dt$ and da/dt if only b is constant?

c. How is dA/dt related to $d\theta/dt$, da/dt, and db/dt if none of a, b, and θ are constant?

20. Heating a plate When a circular plate of metal is heated in an oven, its radius increases at the rate of 0.01 cm/min. At what rate is the plate's area increasing when the radius is 50 cm?

21. Changing dimensions in a rectangle The length l of a rectangle is decreasing at the rate of 2 cm/sec while the width w is increasing at the rate of 2 cm/sec. When $l = 12$ cm and $w = 5$ cm, find the rates of change of **(a)** the area, **(b)** the perimeter, and **(c)** the lengths of the diagonals of the rectangle. Which of these quantities are decreasing, and which are increasing?

